Lebanese University

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Masters in Science and Technology of Information Systems

Advanced Database Research Topic

Adaptive Rules Index Selection for Query Performance Optimization in Large Databases

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Abstract

With the rapid growth of data, database systems must efficiently process large volumes of queries. Indexing plays a crucial role in improving query performance, but selecting the appropriate indexes is a complex and dynamic problem.

This project focuses on adaptive index selection techniques for optimizing query execution in large-scale databases. The proposed approach analyzes query workloads and dynamically recommends or adjusts indexes to enhance performance while minimizing storage and maintenance costs.

A prototype system was developed using PostgreSQL to evaluate different indexing strategies, including B-Tree and Hash indexes. Experimental results demonstrate significant improvements in query execution time when adaptive indexing strategies are applied.

This work highlights the importance of intelligent, workload-aware indexing in modern database systems and provides insights into automated optimization techniques.

Chapter 1: Introduction

1.1 Context

Modern applications generate massive amounts of data, requiring efficient database systems to store and retrieve information. Query performance is a critical factor in system efficiency.

1.2 Problem Statement

Choosing the right indexes manually is difficult due to changing query patterns and large data volumes. Poor index selection can lead to slow performance. Traditional indexing strategies use static index selection and do not adapt to query type. Static indexing is a fixed set of indexes defined at design time. It cannot adapt to changing query patterns or data distribution.

But queries vary, we have:

1- Equality queries

2- Range queries

3- Category filters

PS: Some queries may not use any index → full table scans.

This leads to inefficient execution, increased response time, and resource waste.

1.3 Objectives

- Analyze index impact on performance
- Design adaptive indexing
- Evaluate performance
- Compare strategies

1.4 Methodology

The methodology includes workload analysis, index creation, performance measurement, and adaptive optimization.

Chapter 2: Background & Literature Review

2.1 Database Indexing

Indexes are data structures that improve data retrieval speed. Common types include B-Tree and Hash indexes. A database index is a separate data structure, maintained alongside the main table, that stores a subset of the table's data (typically one or more column values) in a form that supports rapid lookups. An index works similarly to the index at the back of a textbook: instead of reading every page to find a topic, you consult the index to jump directly to the relevant page. In database terms, the index maps column values to the physical locations (page and row offset) of matching records, enabling the query engine to retrieve data without scanning irrelevant rows.

2.2 Query Optimization

Query optimizers choose execution plans based on cost estimation. Indexes reduce query execution time. The performance gap between indexed and non-indexed queries grows dramatically with table size. For a table with N rows, a sequential scan requires O(N) page reads. A well-designed B-Tree index reduces this to O(log N). Consider a table with 500,000 rows: a sequential scan may read thousands of disk pages, while an index scan might read only 3 or 4 index levels before locating the target row.

In practice, however, the relationship between indexing and performance is not straightforward. Indexes introduce overhead on write operations (INSERT, UPDATE, DELETE), consume additional storage space, and must be carefully chosen to match the actual query workload. This is why adaptive index selection, the central theme of this project, is a critical area of database optimization research.

2.3 Related Work

Research explores automated indexing, adaptive systems, and self-tuning databases.

Chapter 3: System Analysis

3.1 Functional Requirements

- Execute queries
- Create/drop indexes
- Measure performance
- Recommend indexes

3.2 Non-Functional Requirements

- Performance
- Scalability
- Reliability

Chapter 4: Database Design

4.1 Database Design

Table simulate real-world datasets with large volumes of data. We used PostgreSQL to create Products table with 500,000 rows of data. And we created different and random data set for each attribute.

	product_id [PK] integer	product_name character varying (150)	category character varying (50)	subcategory character varying (50)	brand character varying (80)	price numeric (10,2)	stock_quantity integer	rating numeric (3,1)	is_available boolean	created_at date	updated_at timestamp without time zone
1	1	Nike Deluxe Station 1001	Electronics	Laptops	Nike	1593.47	494	3.7	true	2019-05-30	2026-02-21 14:14:51.839992
2	2	KitchenAid Wireless Panel 1002	Electronics	Tablets	KitchenAid	1940.51	248	3.5	true	2019-01-12	2026-02-21 14:14:51.839992
3	3	Mattel Portable Bundle 1003	Electronics	Accessories	Mattel	1962.37	153	3.8	true	2019-02-28	2026-02-21 14:14:51.839992
4	4	Johnson & Johnson Smart Unit 1004	Clothing	Men's Wear	Johnson & Johnson	374.71	2433	3.3	true	2022-11-05	2026-02-21 14:14:51.839992
5	5	Puma Smart Edition 1005	Clothing	Women's Wear	Puma	324.92	0	3.9	true	2020-04-07	2026-02-21 14:14:51.839992
6	6	Apple Super Device 1006	Clothing	Kids Wear	Apple	309.94	481	3.3	false	2021-06-22	2026-02-21 14:14:51.839992
7	7	Lenovo Premium Collection 1007	Home & Garden	Furniture	Lenovo	740.21	412	3.2	true	2023-06-07	2026-02-21 14:14:51.839992
8	8	Lenovo Professional Device 1008	Home & Garden	Kitchen	Lenovo	354.04	439	3.7	true	2020-11-09	2026-02-21 14:14:51.839992
9	9	Puma Lite Controller 1009	Home & Garden	Garden Tools	Puma	1095.29	8152	3.8	true	2018-04-11	2026-02-21 14:14:51.839992
10	10	Penguin Turbo Collection 1010	Sports	Fitness	Penguin	300.15	5321	3.9	true	2024-06-22	2026-02-21 14:14:51.839992
11	11	Mattel Professional Kit 1011	Sports	Outdoor	Mattel	834.61	33	3.4	true	2018-05-01	2026-02-21 14:14:51.839992
12	12	Dove Smart Kit 1012	Books	Fiction	Dove	9.80	218	4.1	true	2021-02-27	2026-02-21 14:14:51.839992
13	13	Apple Premium Kit 1013	Books	Non-Fiction	Apple	34.58	31	3.6	true	2021-10-30	2026-02-21 14:14:51.839992
14	14	Random House Max Controller 1014	Beauty	Skincare	Random House	276.67	498	4.1	true	2018-04-18	2026-02-21 14:14:51.839992

Total rows: 500000 Query complete 00:00:01.501 CRLF Ln 1, CRLF Ln 1, Col 1

Figure 1

The attributes are:

```
CREATE TABLE products (
   product_id      SERIAL PRIMARY KEY,
   product_name    VARCHAR(150)       NOT NULL,
   category        VARCHAR(50)        NOT NULL,
   subcategory     VARCHAR(50)        NOT NULL,
   brand           VARCHAR(80)        NOT NULL,
   price           NUMERIC(10, 2)     NOT NULL,
   stock_quantity  INT                NOT NULL,
   rating          NUMERIC(3, 1)      NOT NULL,
   is_available    BOOLEAN            NOT NULL DEFAULT TRUE,
   created_at      DATE               NOT NULL,
   updated_at      TIMESTAMP          NOT NULL DEFAULT NOW()
);
```

Chapter 5: Implementation

5.1 Tools & Technologies

- PostgreSQL
- SQL

5.2 Indexing Strategies and Queries

5.2.1 Querying without Indexes

Step 1: Cardinality recap

Cardinality recap

Column	Approx. distinct values	Cardinality
brand	500	High
category	20	Medium
rating	5	Low
status	3	Low

Step 2: Perform different queries and write down the performance

Q1: Equality on brand (high cardinality):

EXPLAIN (ANALYZE, BUFFERS) **SELECT * FROM products WHERE brand = 'Brand-10';**
Phase 1 – No index (sequential scan) 1. First, drop any existing indexes on brand: DROP INDEX IF EXISTS idx_prod_brand_btree; DROP INDEX IF EXISTS idx_prod_brand_hash; ANALYZE products; 2. Run the query: SET max_parallel_workers_per_gather = 0; EXPLAIN (ANALYZE, BUFFERS) SELECT * FROM products WHERE brand = 'Brand-10';
Phase 2 – B-Tree index 1. Create the B-Tree index on brand: CREATE INDEX idx_prod_brand_btree ON products(brand); ANALYZE products; 2. Run the same query:

```
SET max_parallel_workers_per_gather = 0;
EXPLAIN (ANALYZE, BUFFERS)
SELECT * FROM products WHERE brand = 'Brand-10';
```

Phase 3 – Hash index:

1. Drop the B-Tree index and create the Hash index

```
DROP INDEX idx_prod_brand_btree;
CREATE INDEX idx_prod_brand_hash ON products USING hash (brand);
ANALYZE products;
```

2. Run the same query

```
SET max_parallel_workers_per_gather = 0;
EXPLAIN (ANALYZE, BUFFERS)
SELECT * FROM products WHERE brand = 'Brand-10';
```

Q2: Equality on rating (low cardinality):

SELECT * FROM products WHERE rating = 5;

Phase 1 – No index

```
DROP INDEX IF EXISTS idx_prod_rating_btree;
DROP INDEX IF EXISTS idx_prod_rating_hash;ANALYZE products;
SET max_parallel_workers_per_gather = 0;
EXPLAIN (ANALYZE, BUFFERS)
SELECT * FROM products WHERE rating = 5;
```

Phase 2 – B-Tree index

```
CREATE INDEX idx_prod_rating_btree ON products(rating);ANALYZE products;SET
max_parallel_workers_per_gather = 0;EXPLAIN (ANALYZE, BUFFERS)SELECT * FROM products WHERE
rating = 5;
```

Phase 3 – Hash index

```
DROP INDEX idx_prod_rating_btree;
CREATE INDEX idx_prod_rating_hash ON products USING hash (rating);
ANALYZE products;SET max_parallel_workers_per_gather = 0;
EXPLAIN (ANALYZE, BUFFERS)
SELECT * FROM products WHERE rating = 5;
```

Q3: Equality on category (medium cardinality):

```
SELECT * FROM products WHERE category = 'Category-3';
```

Phase 1 – No index

```
DROP INDEX IF EXISTS idx_prod_category_btree;
DROP INDEX IF EXISTS idx_prod_category_hash;
ANALYZE products;SET max_parallel_workers_per_gather = 0;
EXPLAIN (ANALYZE, BUFFERS)
SELECT * FROM products WHERE category = 'Category-3';
```

Phase 2 – B-Tree index

```
CREATE INDEX idx_prod_category_btree ON products(category);
ANALYZE products;
SET max_parallel_workers_per_gather = 0;
EXPLAIN (ANALYZE, BUFFERS)
SELECT * FROM products WHERE category = 'Category-3';
```

Phase 3 – Hash index

```
DROP INDEX idx_prod_category_btree;
CREATE INDEX idx_prod_category_hash ON products USING hash (category);
ANALYZE products;SET max_parallel_workers_per_gather = 0;
EXPLAIN (ANALYZE, BUFFERS)
SELECT * FROM products WHERE category = 'Category-3';
```

Q4: Range query: price BETWEEN 50 AND 60 (narrow range):

```
SELECT * FROM products WHERE price BETWEEN 50 AND 60;
```

Extra: No Hash in Range Queries

Hash index is not used for range queries:

1. Create a Hash index on price

```
DROP INDEX IF EXISTS idx_prod_price_btree;
CREATE INDEX idx_prod_price_hash ON products USING hash (price);
ANALYZE products;
```

2. Run a range query

```
SET max_parallel_workers_per_gather = 0;
EXPLAIN (ANALYZE, BUFFERS)
SELECT * FROM products WHERE price BETWEEN 50 AND 60;
```

Explanation:

We created a Hash index on price and ran the range query WHERE price BETWEEN 50 AND 60. PostgreSQL did not use the Hash index and chose a sequential scan instead. This shows that Hash indexes do not support range predicates and are only useful for equality lookups.The output will show Seq Scan on products. The Hash index on price will not appear in the plan, because Hash indexes only support equality (=) and not range predicates (BETWEEN, <, >, etc.).

Phase 1 – No index

```
DROP INDEX IF EXISTS idx_prod_price_btree;

ANALYZE products;

SET max_parallel_workers_per_gather = 0;

EXPLAIN (ANALYZE, BUFFERS)

SELECT * FROM products WHERE price BETWEEN 50 AND 60;
```

Phase B – B-Tree index

```
CREATE INDEX idx_prod_price_btree ON products(price);
ANALYZE products;
SET max_parallel_workers_per_gather = 0;
EXPLAIN (ANALYZE, BUFFERS)
SELECT * FROM products WHERE price BETWEEN 50 AND 60;
```

Q5: Aggregation: GROUP BY category with filter (filter is to avid seq scan):

```
SELECT category, AVG(price)
FROM products
WHERE category = 'Category-3'
GROUP BY category;
```

Phase 1 – No index

```
DROP INDEX IF EXISTS idx_prod_category_btree;
DROP INDEX IF EXISTS idx_prod_category_hash;
ANALYZE products;

SET max_parallel_workers_per_gather = 0;

EXPLAIN (ANALYZE, BUFFERS)
SELECT category, AVG(price)
FROM products
WHERE category = 'Category-3'
GROUP BY category;
```

Phase 2 – B-tree index:

```
CREATE INDEX idx_prod_category_btree ON products(category);

ANALYZE products;



SET max_parallel_workers_per_gather = 0;

EXPLAIN (ANALYZE, BUFFERS)

SELECT category, AVG(price)

FROM products

WHERE category = 'Category-3'

GROUP BY category;
```

Phase 3 – Hash index :

```
DROP INDEX idx_prod_category_btree;
CREATE INDEX idx_prod_category_hash ON products USING hash (category);
ANALYZE products;

SET max_parallel_workers_per_gather = 0;

EXPLAIN (ANALYZE, BUFFERS)
SELECT category, AVG(price)
FROM products
WHERE category = 'Category-3'
GROUP BY category;
```

Step 3: Phases tested Results:

Phase 1:

PHASE 1 — Baseline Queries (No Indexes / Sequential Scan)

	Query Description	SQL Pattern	Scan Type (from EXPLAIN)	Planning Time (ms)	Execution Time (ms)	Shared read blocks	Shared hit blocks
1	Equality — brand — high	WHERE brand = 'Brand-10';	Seq Scan on products	0.144	60.716	0	7029
2	Equality — rating — low	WHERE rating = 5	Seq Scan on products	0.985	72.222	0	7029
3	Equality — category — medium	WHERE category = 'Category-3'	Seq Scan on products	0.833	74.879	0	7029
4	Range — price	WHERE price BETWEEN 50 AND 60	Seq Scan on products	1.035	170.810	0	7029
5	Aggregation + GROUP BY	SELECT rating, AVG(price) FROM products GROUP BY rating	Seq Scan on products	1.589	252.639	0	7029
6	Aggregation + GROUP BY + Equality Filter	SELECT category, AVG(price) FROM products WHERE category = 'Category-3' GROUP BY category;	Seq Scan on products	0.205	122.318	0	7029
AVERAGE	Average across all baseline queries			0.799	125.597		

Phase 2:

PHASE 2 — B-Tree Index Results

Best for: Range queries, Sorting, Prefix searches, Composite conditions

	Query Description	SQL Pattern	Scan Type (EXPLAIN)	Planning Time (ms)	Execution Time (ms)	Shared hit	Shared read
1	Equality — brand — high	WHERE brand = 'Brand-10';	Bitmap Index Scan on idx_prod_brand_btree	1.43	1.557	936	4
2	Equality — rating — low	WHERE rating = 5	Bitmap Index Scan on idx_prod_rating_btree	1.998	22.046	7029	56
3	Equality — category — medium	WHERE category = 'Category-3'	Bitmap Index Scan on idx_prod_category_btree	1.379	11.820	6893	25
4	Range — price	WHERE price BETWEEN 50 AND 60	Bitmap Index Scan on idx_prod_price_btree	1.54	4.102	3588	16
5	Aggregation + GROUP BY	SELECT rating, AVG(price) FROM products GROUP BY rating	Seq Scan on products	1.201	292.514	7029	0
6	Aggregation + GROUP BY + Equaltiy Filter	SSELECT category, AVG(price) FROM products WHERE category = 'Category-3' GROUP BY category;	Bitmap Index Scan on idx_prod_category_btree	0.274	41.732	6893	25
AVERAGE	Average across B-Tree queries			1.304	62.295		

Phase 3:

PHASE 3 — Hash Index Results							
Best for: Exact equality searches only — CANNOT support range queries or sorting							
	Query Description	SQL Pattern	Scan Type (EXPLAIN)	Planning Time (ms)	Execution Time (ms)	Shared hit	Shared read
1	Equality — brand — high	WHERE brand = 'Brand-10';	Bitmap Index Scan on idx_prod_brand_hash	1.69	1.859	942	0
2	Equality — rating — low	WHERE rating = 5	Bitmap Index Scan on idx_prod_rating_hash	1.661	23.679	7183	0
3	Equality — category — medium	WHERE category = 'Category-3'	Bitmap Index Scan on idx_prod_category_hash	1.671	15.510	6959	0
4	Range — price (Hash fails)	WHERE price BETWEEN 50 AND 60;	Seq Scan on products	1.859	161.152	7029	0
5	Aggregation + GROUP BY + Equality Filter	SELECT category, AVG(price) FROM products WHERE category = 'Category-3' GROUP BY category;	Bitmap Index Scan on idx_prod_category_hash	0.158	19.121	6959	0
AVERAGE	Average across Hash queries			1.720	50.550		

Step 4: Bitmap Exclusivity

We can test bitmap-based indexing behavior in PostgreSQL that's already seen it (those Bitmap Heap Scan / Bitmap Index Scan plans), but we cannot create a separate BITMAP index type.

The idea is that:

- Index type you create: B-Tree and Hash.
- Access method we test: Bitmap Index Scan (visible in EXPLAIN).

➢ So to "test bitmap-based index":

Create normal B-Tree indexes on **low-cardinality** columns:

```
CREATE INDEX idx_prod_rating_btree ON products(rating);
CREATE INDEX idx_prod_status_btree ON products(status);
ANALYZE products;
```

Use a query that encourages bitmap scans, combined predicates:

```
SET max_parallel_workers_per_gather = 0;
SET enable_bitmapscan = on;
EXPLAIN (ANALYZE, BUFFERS)
SELECT *   FROM products   WHERE rating = 5     AND status = 'ACTIVE;
```

Compare with bitmap disabled to see the difference:

```
SET enable_bitmapscan = off;
EXPLAIN (ANALYZE, BUFFERS)
SELECT *   FROM products   WHERE rating = 5     AND status = 'ACTIVE';
```

➢ "Bitmap-based access" (bitmapscan = on, plan uses Bitmap Index/Heap Scan)
➢ "Non-bitmap index access" (bitmapscan = off, same indexes but different plan)

- Here, we then treat “bitmap indexing” as the bitmap scan access method using B-Tree indexes, and compare its performance against plain Index Scan and Seq Scan, just like you did for B-Tree and Hash.

PHASE 4 — Bitmap Index Results							
Best for: Low-cardinality columns, OR/AND combinations, Aggregations							
Q#	Query Description	SQL Pattern	Scan Type (EXPLAIN)	Planning Time (ms)	Execution Time (ms)	hit	read
Q1	Bitmap scan access method using B-Tree indexes(SET enable_bitmapscan = on;)	SELECT * FROM products WHERE rating = 5 AND status = 'ACTIVE';	Bitmap Index Scan on idx_prod_rating_btree	0.21	35.498	7029	56
Q2	No bitmap scan access method using B-Tree indexes(SET enable_bitmapscan = off;)	SELECT * FROM products WHERE rating = 5 AND status = 'ACTIVE';	Seq Scan on products	0.147	254.458	7029	0
AVERAGE	Average across Bitmap queries			0.179	144.978		

Chapter 6: Final Testing & Results

6.1 Experiment Setup

Queries executed with and without indexes.

6.2 Performance Metrics

- Planning time
- Execution time

6.3 Final Comparative Results

Adaptive indexing improves performance significantly.

FINAL COMPARISON — All Index Types vs Baseline					
PART A — Common Queries: Baseline vs B-Tree vs Hash (Execution Time in ms)					
Query	Baseline (ms)	B-Tree (ms)	Hash (ms)	Best Index	Improvement vs Baseline (%)
Equality — brand — high	60.716	1.557	1.859	B-Tree	97.44%
Equality — rating — low	72.222	22.046	23.679	B-Tree	69.47%
Equality — category — medium	74.879	11.820	15.510	B-Tree	84.21%
Range — price	170.810	4.102	161.152	B-Tree	97.60%
Aggregation + GROUP BY	252.639	292.514	292.514	Baseline	-15.78%
Aggregation + GROUP BY + Equality Filter	122.318	41.732	19.121	Hash	84.37%
AVERAGE	**125.597**	**62.295**	**85.639**		**69.55%**

From these results we conclude **Key Adaptive Rule:**

- IF query_type = 'equality' AND column_cardinality ≤ 20 → USE Bitmap Index
- IF query_type = 'equality' AND column_cardinality > 20 → USE B-Tree Index

- IF query_type = 'range' OR 'ORDER BY' → USE B-Tree Index
- IF query_type = 'OR/AND' AND low_cardinality → USE Bitmap Index
- IF selectivity > 10% OR query_type = 'aggregation' → USE Sequential Scan

Query Type	Condition	Recommended Index	Priority
Equality query	Low cardinality (≤ 20 values)	**Bitmap Index**	High
Equality query	High cardinality (> 20 values)	**B-Tree Index**	High
Range query	Any cardinality	**B-Tree Index**	High
OR / AND combination	Low cardinality columns	**Bitmap Index**	High
Aggregation / GROUP BY	Full table needed	**No index (Seq)**	Medium
Range + ORDER BY	Numeric / Date column	**B-Tree Index**	Medium
Boolean filter	Very low cardinality (2 vals)	**Bitmap Index**	Medium

Chapter 7: Final Conclusion

This study confirms that no single index type universally outperforms all others across every query scenario. The choice of index must be guided by a careful analysis of query characteristics and data properties. B-Tree indexes emerge as the most versatile and reliable choice for general database workloads, capable of handling equality, range, and sorted queries with consistently strong performance. Bitmap indexes offer a specialized but valuable complement for low-cardinality columns and complex Boolean conditions. Hash indexes, while theoretically sound, offer limited practical advantage in systems like PostgreSQL where B-Tree indexes can perform Index Only Scans.

Most significantly, the experiments demonstrate that selectivity is the single most important factor in determining whether any index will be used at all. An adaptive index selection system that accounts for query type, column cardinality, and predicate selectivity can achieve substantial query performance improvements — in some cases exceeding 99% reduction in execution time — while avoiding the pitfall of applying indexes indiscriminately to queries that are better served by sequential scans.

References

Database System Concepts – Silberschatz
PostgreSQL Documentation
Research papers on indexing

Feature	Sequential Scan	Bitmap Scan
Method	Reads the entire table, row by row, in physical order.	Scans index(es) to create a bitmap, then reads only necessary data pages efficiently.
Selectivity	Used for low selectivity (most of the table is needed).	Used for medium selectivity (some rows are needed, but not a small handful or a majority).
Index Usage	Does not use an index to find rows.	Uses one or more indexes to build the in-memory bitmap.
I/O Pattern	Efficient sequential I/O.	Batched, ordered sequential I/O to visit only relevant pages.
Query Plan Node	`Seq Scan`	`Bitmap Index Scan` + `Bitmap Heap Scan`